

$^{37}\text{Cl}(\text{d},\alpha\gamma)$ 1975VaYG,1972Va07

1975VaYG,1972Va07: A 4.25-MeV deuteron beam of 55 nA was provided from the Groningen 5 MV Van de Graaff generator.

The target was a $100\text{ }\mu\text{g}/\text{cm}^2$ Co^{37}Cl enriched to 98% evaporated onto $10\text{--}\mu\text{g}/\text{cm}^2$ Formva plus $10\text{--}\mu\text{g}/\text{cm}^2$ carbon. α particles were detected using a $60\text{--}\mu\text{m}$ annular silicon detector. γ rays were detected using a 120 cm^3 Ge(Li) detector at 90° . Measured $\sigma(E_\alpha)$, E_γ , I_γ , and $\alpha\gamma$ -coin. Deduced levels, γ -ray branching ratios. Also see [1971VaZG](#).

1968Te06: 3.1-4.6-MeV deuteron beams of 50 nA were provided from the Groningen 5-MV Van de Graaff generator. The target was a $100\text{--}\mu\text{g}/\text{cm}^2$ Co^{37}Cl both of natural ^{37}Cl abundance and enriched to 93% evaporated onto $10\text{--}\mu\text{g}/\text{cm}^2$ Formva plus $10\text{--}\mu\text{g}/\text{cm}^2$ carbon. α particles were detected using an annular solid-state detector at $168\text{--}173^\circ$. γ rays were detected using a 3 in. by 3 in. NaI(Tl) scintillator at 55° . Measured $\sigma(E_\alpha)$, E_γ , and $\alpha\gamma$ -coin. Deduced levels.

1955Pa54: $^{37}\text{Cl}(\text{d},\alpha)^{35}\text{S}$ with 3.0, 5.6, 6.0, 7.0, and 7.5-MeV deuteron beams from the MIT-ONR electrostatic generator. Targets were 80^- and $300\text{--}\mu\text{g}/\text{cm}^2$ Barium chloride (75.4% ^{35}Cl , 24.6% ^{37}Cl) evaporated onto formvar films on a gold layer. Charged reaction products emitted at 90° were magnetically analyzed by a broad-range spectrograph. Measured $\sigma(E_\alpha)$. Deduced levels.

 ^{35}S Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	Comments
0	$3/2^+$	
1572.2 12	$1/2^+$	
1990.0 10	$5/2^-, 7/2^-$	E(level): Other: 1992 10 from 1955Pa54 .
2348.2 10	$3/2^-$	E(level): Other: 2348 10 from 1955Pa54 .
2716.7 10	$(3/2, 5/2, 7/2)$	E(level): Other: 2714 10 from 1955Pa54 .
2939.2 13	$(3/2, 5/2)$	
3423 5		
3560.8 19		
3598.4 21		
3803.6 19		
3818.1 11		
3889.0 19		
4022.2 22		
4027.7 10		E(level): Other: 4025 10 from 1955Pa54 .
4108 3		
4180 3		
4187 3		
4302 4		
4480.0 16		

† From [1975VaYG](#) based on γ -ray energies.

‡ As given in [1972Va07](#).

 $\gamma(^{35}\text{S})$

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
1572.2	$1/2^+$	1572.2	100	0	$3/2^+$	
1990.0	$5/2^-, 7/2^-$	1989.9	100	0	$3/2^+$	
2348.2	$3/2^-$	776.0	25 2	1572.2	$1/2^+$	
		2348.1	75 2	0	$3/2^+$	
2716.7	$(3/2, 5/2, 7/2)$	2716.6	100	0	$3/2^+$	
2939.2	$(3/2, 5/2)$	2939.1	100	0	$3/2^+$	
3423		3423	100	0	$3/2^+$	
3560.8		1212.6	35 4	2348.2	$3/2^-$	
		1570.8	65 4	1990.0	$5/2^-, 7/2^-$	
3598.4		3598.2	100	0	$3/2^+$	I_γ : >95 in 1975VaYG .
3803.6		2231.3	38 3	1572.2	$1/2^+$	
		3803.4	62 3	0	$3/2^+$	
3818.1		1828.1	100	1990.0	$5/2^-, 7/2^-$	
3889.0		1540.8	40 4	2348.2	$3/2^-$	

Continued on next page (footnotes at end of table)

$^{37}\text{Cl}(\text{d},\alpha\gamma)$ 1975VaYG,1972Va07 (continued) $\gamma(^{35}\text{S})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
3889.0	1898.9	45 5	1990.0	$5/2^-, 7/2^-$	
	3888.8	15 3	0	$3/2^+$	
4022.2	2032.1	100	1990.0	$5/2^-, 7/2^-$	
4027.7	1088.5	33 4	2939.2	$(3/2, 5/2)$	
	1679.5	33 4	2348.2	$3/2^-$	
	2455.4	34 6	1572.2	$1/2^+$	
4108	4108	100	0	$3/2^+$	I_γ : >95 in 1975VaYG.
4180	2608	18 5	1572.2	$1/2^+$	
	4180	82 5	0	$3/2^+$	
4187	1839	100	2348.2	$3/2^-$	I_γ : >95 in 1975VaYG.
4302	1954	59 5	2348.2	$3/2^-$	
	4302	41 5	0	$3/2^+$	
4480.0	1763.3	36 4	2716.7	$(3/2, 5/2, 7/2)$	
	2489.9	15 4	1990.0	$5/2^-, 7/2^-$	
	2907.7	39 4	1572.2	$1/2^+$	
	4479.7	10 2	0	$3/2^+$	

† Deduced by evaluators from level-energy difference in 1975VaYG.

‡ From 1975VaYG.

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Level Scheme

Intensities: % photon branching from each level

